

REVIEW

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Exploring drug-induced liver injury: comprehensive insights into mechanisms and management of hepatotoxic agents

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Abstract

Background Drug-induced liver injury (DILI) is a significant adverse drug reaction, manifesting through a range of clinical presentations from mild liver enzyme to acute liver failure.

Main text This review provides a comprehensive overview of DILI, emphasizing the differences between intrinsic and idiosyncratic DILI. The underlying molecular mechanisms, like mitochondrial dysfunction, oxidative stress, and immune-mediated responses, are discussed in detail. The epidemiology of DILI is explored through various retrospective and prospective studies, highlighting the role of specific medications and individual susceptibility factors. The review also addresses the challenges in diagnosing DILI and the impact on drug development and clinical practice.

Conclusion DILI poses a significant clinical threat due to its potential for causing acute liver failure and associated mortality. To improve patient outcomes, further research is crucial to identify effective therapeutic interventions.

Keywords Drug-induced liver injury, DILI, Hepatotoxicity, Intrinsic DILI, Idiosyncratic DILI, Oxidative stress, Mitochondrial dysfunction, Immune-mediated DILI, Epidemiology, And drug safety

Introduction

Drug-induced liver injury (DILI) is a serious adverse drug reaction where various medications, including prescription drugs, supplements, herbal remedies, and illegal substances, damage the liver [1]. This harm occurs because the liver metabolizes xenobiotics entering the body through the digestive tract. Despite affecting only an estimated 14–19 cases per 100,000 people, DILI is a significant contributor to acute liver injury (ALI). Drug-induced liver injury is a significant challenge in clinical settings due to its association with high mortality rates, particularly in cases of acute liver failure [2].

Classification of drug-induced liver injury (DILI)

Intrinsic DILI

Certain drugs and chemicals have inherent liver toxicity, causing a predictable spectrum of liver damage that worsens with increase in dose or exposure [3]. This type of injury, intrinsic hepatotoxicity, typically occurs soon after exceeding a threshold dose, which can vary between individuals. Drugs that trigger intrinsic DILI are frequently lipophilic, allowing them to pass through the liver cell's fatty membrane easily. These drugs are metabolized into harmful by-products that disrupt the bile acid balance and damage the cell's energy production machinery through oxidative stress and cellular signaling pathways [4].

A prime example of intrinsic DILI is acetaminophen poisoning, also known as N Acetyl P-aminophenol (APAP) or paracetamol poisoning. In the USA and some European countries, it is responsible for roughly half of all acute liver failure (ALF) cases. A significant number

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of people unintentionally develop liver damage from acetaminophen, even at doses just exceeding the recommended daily limit of 4 g. This has been termed 'therapeutic misadventure' because it happens with doses intended for treatment. It points out that even healthy people can experience liver damage (increased ALT) with repeated high doses. Several factors, like fasting, alcohol, other medications, and existing illnesses, can make people more susceptible to acetaminophen's toxic effects at lower doses [5].

Idiosyncratic DILI

On the other hand, most drugs that cause DILI have a much lower and unpredictable risk. This type, called idiosyncratic DILI, is linked to specific genetic or other factors that make some people more susceptible. Idiosyncratic DILI is a rare liver reaction to medications, affecting fewer than 1 in 10,000 people. It typically develops after a longer delay, ranging from days to weeks, and cannot be predicted. It was believed that the body's adaptive immune system and the human leukocyte antigens (HLA) might be crucial factors in causing idiosyncratic DILI in people with a genetic predisposition [6].

Idiosyncratic DILI can be categorized into immune-mediated (allergic) and nonimmune-mediated reactions. Immune-mediated idiosyncratic DILI typically develops within 1 to 6 weeks of starting a medication. Symptoms often include fever, rash, and increased white blood cells, especially eosinophils. In some cases, people may also develop autoantibodies (e.g., anti-smooth-muscle antibody or antinuclear antibody) that attack the body's tissues [7]. Nonimmune-mediated idiosyncratic DILI is a type of DILI not caused by the immune system. It is a rare and unpredictable side effect of certain medications. The precise mechanisms underlying nonimmune-mediated idiosyncratic drug-induced liver injury (DILI) remain unclear. However, it is believed to result from a combination of factors, including mitochondrial dysfunction, inhibition of bile salt export pumps, and genetic predisposition [8] (Table 1). In addition to intrinsic and idiosyncratic DILI, medications can also cause liver injury

through various patterns, including steatohepatitis and neoplastic (anabolic steroid-induced or nodular regenerative hyperplasia) and vasculitis (azathioprine-induced hepatic sinusoidal obstruction syndrome) (Fig. 1) [9].

Methodology

We performed a MEDLINE/Google scholar/PubMed search using the terms 'drug-induced liver injury,' 'DILI,' and 'hepatotoxicity, Recent findings in DILI, management of DILI' and analyzed recent literature to identify relevant articles. We found that these specific 'ahead of print' reports provided valuable insights and decided to include them in this review. We tried to include articles published between 2020 and 2024. Older studies were occasionally cited to provide context for newer findings. We focused on reporting new findings on established hepatotoxins, previously unidentified hepatotoxins, mechanisms of DILI, and updates from international DILI registries. Our review primarily emphasized human studies, except for preclinical studies presenting groundbreaking insights into DILI research. We primarily excluded individual case reports, abstracts, and review articles when the extent of causality assessment in the source materials was unclear. A significant number of studies included individual case reports and case series, underscoring the importance of the Roussel Uclaf Causality Assessment Method (RUCAM) as our primary tool for identifying DILI attributed to specific agents.

Results and discussion

Epidemiology

Accurately determining the incidence of idiosyncratic DILI is a significant challenge. This difficulty is primarily due to the complexities of diagnosis, which can be delayed or missed entirely, leading to underreporting of actual cases. The data on DILI's epidemiology came primarily from retrospective studies conducted by pharmaceutical companies, pharmacovigilance centers, and general practitioners. These studies have concentrated on pinpointing the medications associated with DILI,

Table 1 Comparison of Intrinsic and Idiosyncratic DILI [6, 10, 11]

Characteristics	Intrinsic DILI	Idiosyncratic DILI
Common drugs involved	Acetaminophen (APAP), certain chemotherapy drugs	Antibiotics (e.g., amoxicillin–clavulanate), NSAIDs
Onset time	Soon after exceeding a threshold dose	Unpredictable, varies among individuals
Mechanisms	Predictable toxicity, dose-dependent	Unpredictable, not dose-dependent, immune-mediated responses
Clinical features	Acute liver failure increased ALT levels	Varies widely, and can range from mild to severe liver injury
Severity	High	Variable

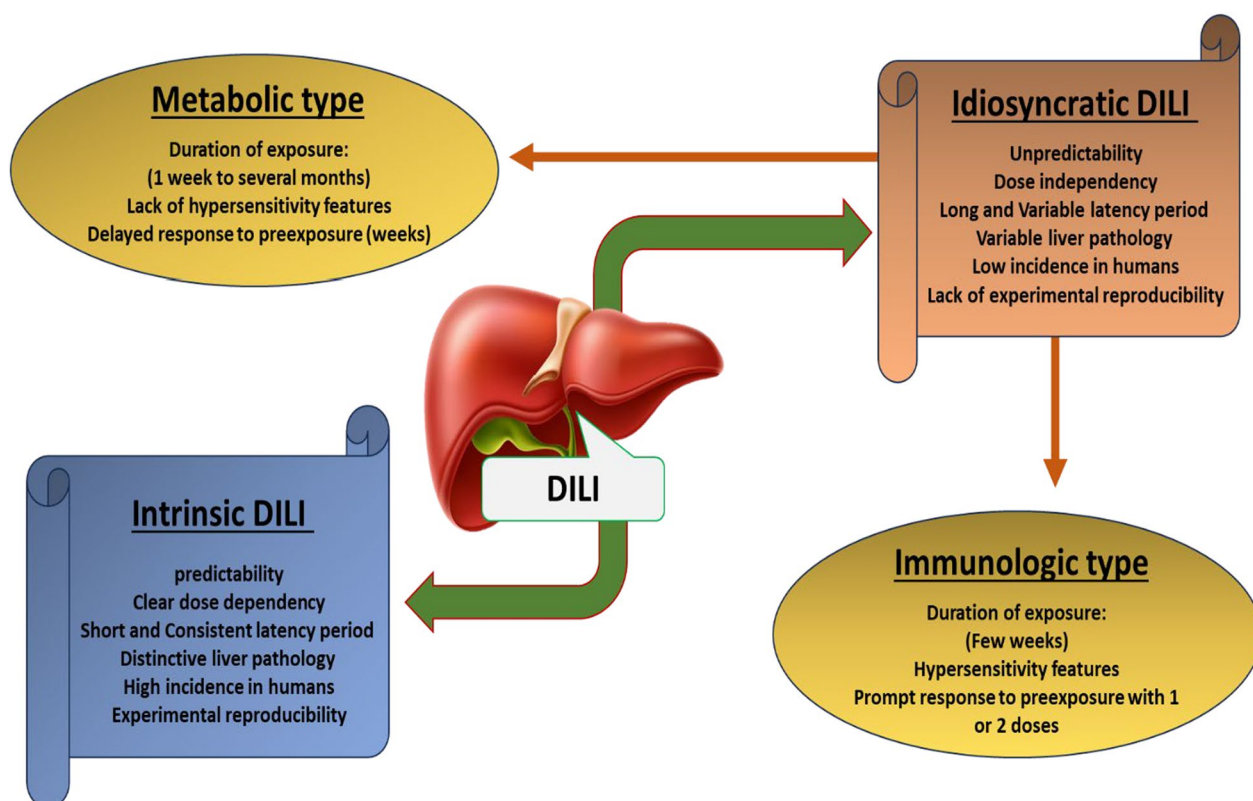


Fig. 1 Characteristics of idiosyncratic DILI and intrinsic DILI

the symptoms experienced by patients, and the liver's long-term impact [4].

A Chinese retrospective study revealed a significantly higher DILI rate than Western countries. It was estimated that the annual incidence was 23.8 cases per 100,000 individuals, significantly surpassing the numbers reported in other regions [12]. A UK study analyzing a database of prescribed drugs found varying risks of DILI. Ampicillin, NSAIDs, and methyldopa were linked to a low incidence (1 case per 100,000 prescriptions), while erythromycin had a higher risk (14 cases per 100,000 prescriptions). The study also indicates a heightened risk of DILI when more than two medications are used simultaneously [13].

A large, well-designed study in France (prospective and population-based) estimated an annual rate of DILI of around 14 cases per 100,000 people. This is slightly lower than a similar study in Iceland, which found a rate of 19 cases per 100,000 people [14]. Research conducted by Goldberg, D. S. et al. examined the occurrence of drug-induced ALF within Kaiser Permanente, a major integrated healthcare network in Northern California [15]. Another epidemiological study in different regions is mentioned in (Table 2).

Molecular mechanisms underlying DIL

The liver is crucial for drug clearance, especially lipophilic compounds, making it prone to drug-induced toxicity. However, liver injury from xenobiotics varies due to genetic and individual factors, making adverse reactions unpredictable [16]. Drugs enter liver cells via diffusion or transport proteins and undergo metabolism in two phases. Phase I introduces minor structural changes that can significantly affect drug activity. Phase II conjugates endogenous molecules (e.g., glucuronic acid) to enhance water solubility, rendering the drug inactive [17]. Metabolic by-products can be chemically reactive, binding to proteins, DNA, and enzymes, leading to functional changes or triggering immune responses by forming immunogenic haptens. In addition, they can also deplete the cell's antioxidants and disrupt energy production in the mitochondria (Fig. 2) [18].

Induction of oxidative stress

The initial stage in the development of DILI is thought to involve the exposure of hepatocytes to cellular stress. This stress arises from the body's efforts to metabolize drugs, sometimes resulting in the formation of harmful reactive metabolites. The release of these metabolites

Table 2 Epidemiological studies of drug-induced liver injury (DILI)

Study location	Study design	Population	DILI incidence/prevalence	Specific DILI type	Key findings/Notes
China	Retrospective	General population	23.8 cases per 100,000 individuals	Not specified	Significantly higher than Western countries
UK	Retrospective	Prescribed drug users	Ampicillin, NSAIDs, methyldopa: low; erythromycin: high	Not specified	Higher risk with multiple medications
Sweden	Retrospective	Outpatients with liver disease	6.6% had possible DILI	Not specified	
France	Prospective, population-based	General population	~ 14 cases per 100,000 people	Not specified	Slightly lower than Iceland
Iceland	Prospective, population-based	General population	19 cases per 100,000 people	Not specified	
Northern California (Kaiser Permanente)	Retrospective	Integrated healthcare network	1.02 cases per 1,000,000 person-years	Idiosyncratic DILI-induced ALF	Acetaminophen responsible for 46% of ALF cases
Mainland China	Retrospective cohort	General population	25,927 cases (largest cohort)	Not specified	Traditional Chinese medicine and HDS were primary contributors

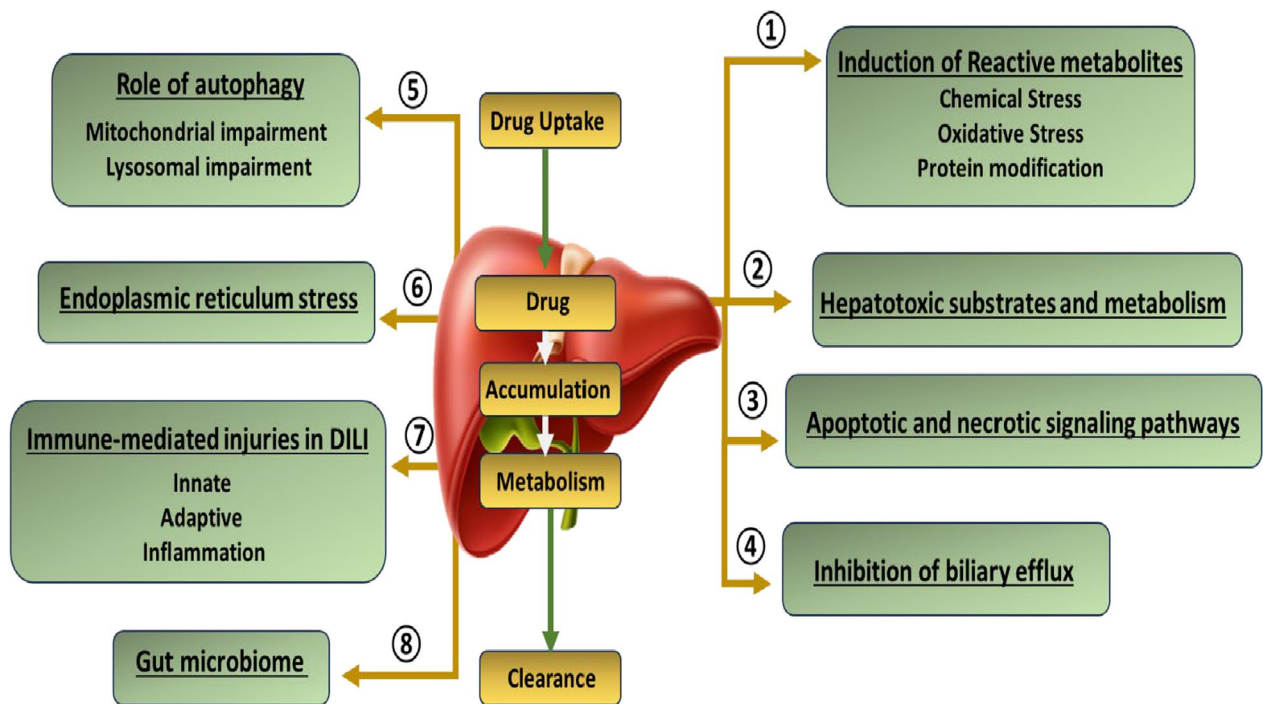


Fig. 2 Molecular mechanisms underlying DILI

can disrupt mitochondrial electron transport complexes, leading to increased production of ROS as a result of premature electron reactions with molecular oxygen [19].

Certain drugs, like cyclophosphamide, undergo redox cycling, generating harmful superoxide radicals. Glutathione (GSH), a tripeptide, plays a crucial role in detoxification, cell growth, apoptosis, structural

organization, and immune function beyond its antioxidant properties [20]. Glutathione (GSH) plays a main role in neutralizing ROS through reduction mechanisms [21]. Excessive intake of acetaminophen results in the formation of a toxic metabolite, NAPQI, which depletes GSH reserves. This depletion allows NAPQI to damage cells and indirectly trigger ROS production

within hepatocytes, potentially leading to liver injury [22].

Inflammation triggers redox stress and mitochondrial dysfunction through pro-inflammatory mediators (IL-6, TNF- α , IL-8, IL-1 β , CXCL10), leading to ATP depletion, increased ROS production, and hepatic damage. Elevated ROS levels activate the mitochondrial permeability transition pore (mPTP), further disrupting mitochondrial function and creating a cycle of oxidative stress [23]. A study on APAP-induced hepatotoxicity found that MAP kinases, particularly JNK, activate mPTP via protein phosphorylation, increasing membrane permeability. JNK plays a key role in linking physiological and pathological states, and its inhibition has been shown to reduce oxidative stress and prevent hepatic injury [24].

Hepatotoxic substrates and metabolism

Drugs are absorbed from the sinusoidal blood into hepatocytes via transporters located in the basolateral membrane. These transporters belong to the families of organic anion transporters (OAT), organic anion transporting polypeptides (OATP), and organic cation transporters (OCT) [25]. Subsequently, after undergoing phase I and phase II metabolism, drugs are eliminated through Phase III. Phase III of drug metabolism involves the active elimination of metabolized drugs via specialized protein pumps. These ATP-dependent efflux pumps, such as MDR, BSEP, and MATE, transport drugs into the sinusoidal blood or bile for excretion [26]. Consequently, the inhibition of drug efflux transporters may cause the buildup of potentially harmful metabolites in hepatocytes, which can contribute to hepatotoxicity. On the other hand, inhibiting basolateral uptake transporters is less likely to cause direct liver damage. However, it can impact the pharmacokinetics of other drugs that utilize the same transporter for uptake [25].

For instance, administering statins alongside OATP1B1 inhibitors like gemfibrozil (a lipid-lowering agent) or cyclosporine A (an immunosuppressant) can elevate statin serum levels, thereby increasing the risk of statin-induced myopathy [27]. Furthermore, genetic variations in multidrug resistance-associated protein 2 (MRP2) have been associated with DILI, because it impairs the ability of hepatocytes to eliminate conjugated anionic drug metabolites [28].

Apoptotic and necrotic signaling pathways Multiple studies have highlighted the role of apoptosis in the pathogenesis of drug-induced liver injury (DILI). This can occur due to direct drug effects on organelles or hepatocyte membrane damage, or activation of signaling stress that ultimately activates apoptotic cascades [29]. Apoptosis occurs via two primary pathways: the extrinsic path-

way, which involves death receptor signaling, and the intrinsic pathway, which is mediated by mitochondria. In the context of DILI, apoptosis is initiated through the intrinsic pathway due to the generation of ROS, leading to a cascade of events that ultimately results in mitochondrial outer membrane permeabilization (MOMP) [30].

Apoptosis is regulated by the balance between pro-apoptotic proteins (Bax, Bak) and anti-apoptotic proteins (Bcl-2), with p53 promoting pro-apoptotic gene expression [31]. In the intrinsic pathway, mitochondrial membrane permeabilization allows cytochrome c to enter the cytosol, forming the apoptosome with APAF-1 and caspase-9, which then activates caspase-3, leading to nuclear fragmentation, protein degradation, and chromatin condensation [32]. The extrinsic pathway is triggered by ligand binding (e.g., FAS, TNF- α , TRAIL) to their receptors, activating FADD, which in turn activates caspase-8 and caspase-10, ultimately leading to apoptosis via caspase-3 and caspase-7 [33].

Necrosis is triggered by a process called mitochondrial permeability transition (MPT). Mitochondrial permeability transition (MPT) disrupts the function of mitochondria, leading to loss of cell membrane integrity, cell swelling, and ultimately cell death. Necrosis, like apoptosis, can be triggered by various factors, including cellular stress (especially oxidative stress), ATP depletion, calpain activation, and even death receptor activation (TNF- α and Fas) [34]. In contrast to acetaminophen, which primarily induces cell death through necrosis, the relative roles of necrosis and apoptosis in DILI remain uncertain for most other medications [35].

Bile duct injury Cholestatic and mixed liver injuries can damage the bile ducts, characterized by the destruction of the bile duct lining epithelium, potentially contributing to the progression of biliary cirrhosis. Obstructions in the bile ducts disrupt normal bile flow, causing bile components to back up and leak into the bloodstream. T cell-mediated hypersensitivity has been documented as a possible cause of bile duct injury, but direct damage from chemically reactive metabolites released during drug metabolism may also play a role [36].

Another potential DILI mechanism involves reducing the bile salt export pump (BSEP). This inhibition results in elevated intracellular concentrations of bile salts, which can cause hepatocellular stress and liver damage [37]. An example of a BSEP inhibitor is bosentan, a drug prescribed for pulmonary hypertension that includes a boxed warning for hepatotoxicity. Furthermore, bile acids can induce hepatocyte apoptosis by enhancing the localization of death receptors to the plasma membrane [38].

Troglitazone, a drug from the thiazolidinedione class, was removed from the market due to significant

hepatotoxicity. The mechanisms underlying its hepatotoxic effects involve mitochondrial dysfunction, oxidative stress, and the inhibition of the bile salt export pump, resulting in the accumulation of toxic bile salts within hepatocytes [39]. Recent findings also indicate that troglitazone inhibits bile acid amidation, resulting in the buildup of non-amidated bile acids, further disrupting bile acid homeostasis and increasing stress on hepatocytes [39].

Role of autophagy Cells have a cleaning process called autophagy where they break down and reuse their parts. This can be done in two ways: general cleaning (nonselective autophagy) during starvation or targeted cleaning (selective autophagy) to get rid of damaged parts like mitochondria or invading bacteria [40]. Autophagy is essential in drug-induced acute liver injuries. When properly regulated, autophagy maintains liver physiology and balances liver metabolism by ensuring cytoplasmic quality through the removal of misfolded proteins and damaged organelles, including mitochondria [41]. When hepatic autophagy is impaired, it may lead to several harmful consequences. First, it can prevent the removal of damaged organelles and proteins, which can lead to hepatic death. Second, it can increase the production of ROS, which can further damage hepatocytes. Third, it can trigger inflammation, which can also contribute to hepatocyte death [42].

Hepatotoxic agents like acetaminophen and alcohol disrupt autophagy, leading to ROS generation, inflammation, and apoptosis in hepatocytes. Increased ROS, such as superoxide and peroxynitrite, drive inflammatory responses and contribute to mitochondrial dysfunction by damaging essential biomolecules [43]. Autophagy plays a crucial role in maintaining mitochondrial quality by degrading damaged organelles, thereby supporting cell survival [44].

Apostolova et al. identified mitochondrial dysfunction as a significant mechanism behind efavirenz-induced liver injury. Further investigations by the same group revealed that efavirenz stimulates autophagy, indicated by the conversion of LC3-I to LC3-II and the autophagic degradation of mitochondria in human hepatic cells. Moreover, co-treatment with the autophagy inhibitor 3-MA heightened the toxicity of efavirenz, suggesting that the autophagy induced by efavirenz serves as a protective mechanism that promotes cell survival [45].

Endoplasmic reticulum stress Endoplasmic reticulum (ER) stress is a key factor in the development of DILI. The ER is vital for protein folding, lipid biosynthesis, and calcium storage. When its function is disrupted, it results in ER stress, which triggers a cellular defense mechanism

known as the unfolded protein response (UPR) [46]. Furthermore, ER stress is acknowledged as a secondary consequence of metabolic dysfunction, frequently caused by the excessive buildup of reactive oxygen species (ROS) and disruptions in intracellular calcium levels. The unfolded protein response (UPR) alleviates ER stress through negative feedback mechanisms, which include degrading mRNA, decreasing protein synthesis, inhibiting translation, and posttranslational processes, and facilitating the transport of misfolded proteins to the cytoplasm for degradation [47].

Many drugs and their metabolites can cause protein misfolding within the ER. This accumulation of misfolded proteins initiates the UPR to restore normal function. However, prolonged ER stress can overwhelm the UPR, so lead to cell death [48]. The UPR is triggered to enhance the expression of genes responsible for protein folding and quality control. This response involves three primary pathways: PKR-like ER kinase (PERK), inositol-requiring enzyme 1 alpha (IRE1 α), and activating transcription factor 6 (ATF6), all of which are activated under conditions of ER stress [49]. Drugs, such as acetaminophen, antibiotics (e.g., amoxicillin-clavulanate), and doxorubicin, can induce ER stress by generating ROS and disrupting protein folding, thereby contributing to liver injury [50].

Immune-mediated injuries in DILI The immune system is vital for both the initiation and recovery from DILI because the liver is populated with a diverse array of immune cells that contribute to both innate and adaptive immune responses. Within the liver, there are several resident innate immune cells such as Kupffer cells, dendritic cells, natural killer cells, innate lymphoid cells, and mucosal-associated invariant T cells. In addition, adaptive immune cells, including T cells, regulatory T cells, cytotoxic T lymphocytes, and helper T cells, are also present [51]. Given its function in filtering biological substances, the liver is frequently exposed to foreign antigens and plays a crucial role in maintaining immune tolerance to prevent excessive responses to these antigens. However, in cases of DILI, hepatic tolerance is often compromised, leading to dysregulated immune responses [52]. Drug metabolites can generate hapten-like drug-protein adducts by binding covalently or non-covalently to CYP450 enzymes. These adducts may then attach to endogenous proteins, forming neoantigen structures that are presented to CD8+ cytotoxic T cells through MHC II complexes. Additionally, pro-inflammatory cytokines, such as Fas and TNF- α , released by CD4+ helper T cells and macrophages, amplify inflammation and apoptosis, further contributing to hepatic injury [53].

Certain drugs, including acetaminophen, isoniazid, and diclofenac, can induce liver injury through

immune-mediated mechanisms by releasing DAMPs. These DAMPs bind to pattern-recognition receptors on antigen-presenting cells, which in turn activate the adaptive immune response. Mechanistically, DAMPs interact with toll-like receptors (TLRs) and purinergic receptors on Kupffer cells, resulting in the secretion of pro-inflammatory mediators such as IL-6, IL-18, IFN- γ , TNF- α , and Fas ligands, which further promote macrophage proliferation [54]. Another mechanism involves the misdirection of endogenous peptides, resulting in HLA autoimmunity and subsequent cell injury [55].

Gut microbiome The gut microbiota is a complex community of microorganisms in the digestive system that aids digestion, nutrient metabolism, immune function, and protection against pathogens. Imbalances in the microbiota are linked to various health issues, including digestive disorders, obesity, immune diseases, and mental health problems [56]. In drug-induced liver injury (DILI), the microbiota affects immune responses and drug metabolism by altering the bioavailability and toxicity of drugs through various biotransformation processes such as hydrolysis, demethylation, and oxidation.[57].

Bacterial translocation typically occurs in small quantities and is regulated by liver immune cells that uphold immune tolerance toward the components of the commensal microbiota. However, disturbances in the balance and composition of the gut microbiota, known as dysbiosis, can enhance intestinal permeability. This increase allows for greater microbial translocation to the liver, exposing hepatic immune cells to foreign antigens and subsequently triggering protective immune responses [58]. Microbial components detected by toll-like receptors on hepatic immune cells can disrupt immune homeostasis by modifying cytokine production or affecting immune cell proliferation. Consequently, dysbiosis may compromise

hepatic immune tolerance, potentially initiating an adaptive immune response. This mechanism could play a role in the development of DILI in individuals taking medications that produce drug-related neoantigens, triggering immune-mediated hepatic injury and compromising liver function [57] (Table 3).

Clinical manifestations and diagnosis

Most cases of DILI are asymptomatic or present with mild, nonspecific symptoms that typically resolve once the offending drug is discontinued. However, this can delay clinical intervention, as the symptoms often resemble other hepatic conditions like autoimmune hepatitis. Therefore, diagnosing DILI usually requires ruling out other causes. In some cases, severe DILI can progress to acute liver failure or develop into chronic liver disease [59]. The severity of liver injury cannot be determined solely by elevated liver enzymes; it also requires consideration of clinical symptoms and other laboratory values. Common symptoms include dark urine, fever, nausea, pruritus, jaundice, vomiting, and pain in the right upper quadrant of the abdomen. The diverse clinical presentations were further presented in (Fig. 3).

Diagnosing DILI involves ruling out other causes by taking a thorough medical history and assessing factors like alcohol abuse, sepsis, congestive heart failure, and ischemic hepatitis. Comorbidities, viral hepatitis risks, and local infections should also be considered. The liver injury pattern helps guide further testing such as ruling out primary biliary cholangitis or sclerosing cholangitis in cases of cholestasis or evaluating for biliary tract obstructions if jaundice is present [59]. While a liver biopsy is typically not necessary for suspected DILI, it is indicated when autoimmune features are present, as it can offer valuable diagnostic information. Research shows that hepatocellular cholestasis and the presence

Table 3 Hepatotoxic drugs and potential molecular mechanisms

Drug class	Specific drug(s)	Primary hepatotoxic pattern	Potential molecular mechanisms	Notes/additional information
Antibiotics	Amoxicillin–clavulanate, isoniazid, fluoroquinolones	Hepatocellular, cholestatic, mixed	Reactive metabolites, immune-mediated responses	Wide range of mechanisms; dose-dependent and idiosyncratic reactions possible
NSAIDs	Diclofenac, ibuprofen, naproxen	Hepatocellular	Oxidative stress, mitochondrial dysfunction	Long-term use often associated with injury
Analgesics/antipyretics	Acetaminophen	Hepatocellular, severe liver failure	NAPQI production, oxidative stress, GSH depletion	Overdose leads to significant liver damage
Statins	Atorvastatin, simvastatin, rosuvastatin	Hepatocellular	Immune-mediated responses, mitochondrial dysfunction	Variable risk and severity of liver injury
Antituberculosis	Isoniazid, pyrazinamide, rifampicin	Hepatocellular	Reactive metabolites, immune-mediated responses	Significant liver toxicity



Fig. 3 Diverse clinical presentations of DILI

of portal neutrophils are indicative of DILI, while fibrosis is more suggestive of autoimmune hepatitis (AIH). (Table 4) [60].

Causality assessment tool

One commonly used clinical assessment method for evaluating the likelihood that a drug is responsible for liver injury, based on clinical and laboratory, is the Rousel Uclaf Causality Assessment Method (RUCAM). The method begins with calculating the R ratio to classify the type of injury as hepatocellular, cholestatic, or mixed, based on the elevations in serum enzyme activity. The R ratio is determined using the formula: $R = (\text{ALT value} / \text{ALT ULN}) / (\text{ALP value} / \text{ALP ULN})$. An R ratio greater than 5 indicates hepatocellular injury, a ratio between 2 and 5 signifies mixed injury and a ratio less than 2 indicates cholestatic injury. (Table 5) [61].

Clinical reports indicate that RUCAM effectively identifies potential causative agents in situations involving multiple drug administrations. It addresses the unpredictability associated with idiosyncratic liver events and emphasizes the absence of diagnostic biomarkers, highlighting the persistent challenges faced in the clinical recognition of DILI [64].

The RUCAM model, endorsed by the Council for International Organizations of Medical Sciences (CIOMS), is now the sole validated tool used by regulators, the pharmaceutical industry, and clinicians to assess causality

Table 4 Diagnostic tools and imaging for DILI

Diagnostic tool	Purpose	Examples/details
Medical history	Rule out alternative causes	Alcohol use, medication history, comorbidities
Liver function tests	Assess liver injury	ALT, AST, ALP, bilirubin
Autoantibody testing	Diagnose autoimmune hepatitis	ANA, ASMA, IgG levels
Liver imaging	Rule out biliary obstruction, focal lesions	Abdominal ultrasound, MRCP, CT
Liver biopsy	Confirm diagnosis, distinguish from AIH	Histological examination

ALT Alanine aminotransferase, *AST* Aspartate aminotransferase, *ALP* Alkaline phosphatase, *AIH* Autoimmune hepatitis, *ANA* Antinuclear antibodies, *ASMA* Anti-smooth muscle antibodies, *IgG* Immunoglobulin G, *MRCP* Magnetic resonance cholangiopancreatography & *CT* Computed tomography

Table 5 Patterns of liver injury and differential diagnosis in DILI [62, 63]

Pattern of injury	Description	Criteria for diagnosis	Condition	Diagnostic considerations
Hepatocellular	Injury primarily affecting liver cells	ALT or AST increased > fivefold above ULN	Alcohol abuse	History of alcohol consumption, elevated liver enzymes
			Autoimmune hepatitis	Positive autoantibodies (ANA, ASMA), elevated IgG levels
			Viral hepatitis	Positive viral hepatitis markers
			Ischemic hepatitis	History of hypotension or shock, elevated liver enzymes
Cholestatic	Injury affecting bile ducts	ALP increased > twofold above ULN	Primary biliary cholangitis	Elevated ALP, positive anti-mitochondrial antibodies
			Sclerosing cholangitis	Imaging showing bile duct strictures
Mixed	Combination of hepatocellular and cholestatic	ALT/ALP ratio between 2 and 5	Sepsis	Signs of infection, positive blood cultures
			Congestive heart failure	Symptoms of heart failure, echocardiography findings

in DILI. It evaluates seven key factors: timing of drug exposure, improvement after stopping the drug, patient risk factors, exclusion of other liver diseases, known drug hepatotoxicity, recurrence after re-exposure, and interactions with other medications (Table 6). It classifies DILI as ‘definite,’ ‘highly probable,’ ‘probable,’ ‘possible,’ ‘unlikely,’ or ‘excluded.’ While the scale standardizes assessments, it should not be applied indiscriminately, as it does not establish causality and may lead to biased conclusions in poorly documented cases. It is intended to support, rather than replace, clinical judgment [65].

The CIOMS/RUCAM scale, while clinically useful, has significant limitations (Table 7). It is complex, includes unclear definitions, and lacks strong supporting data. Its reliance on rechallenge data makes it difficult to achieve high probability classifications without such information. Additionally, patients with preexisting liver conditions may receive lower scores, leading to potential variability in results. The scale’s application to drug-induced liver injury (DILI) from herbal and dietary supplements is further complicated by ingredient inaccuracies and limited product information [67].

LiverTox is an extensive and interactive online resource that offers evidence-based information on liver injuries caused by drugs, dietary supplements, and herbal products. Despite its comprehensive nature, its effectiveness in predicting idiosyncratic DILI has faced scrutiny. Concerns have been raised regarding data quality, the lack of

evidence-based case studies, and the absence of standardized definitions and categorizations of liver injuries, which may limit its reliability and applicability in clinical practice [69].

Emerging biomarkers of hepatic injury

Biomarkers of DILI are crucial for early detection, diagnosis, and monitoring of liver damage caused by medications. These biomarkers can be categorized into traditional biochemical markers and emerging molecular biomarkers. Traditional serum liver tests, including AST, ALT, ALP, and bilirubin, have been utilized in clinical practice and drug development since the 1950s to identify and monitor DILI. However, these biomarkers have several limitations: they are not liver-specific, they do not reliably predict DILI outcomes, and they provide limited mechanistic insights into the underlying processes of liver injury. These shortcomings underscore the need for more precise biomarkers in the assessment of DILI. Therefore, it is prompted to discover and validate new biomarkers [70]. One promising biomarker for liver injury is miR-122, a small RNA molecule that constitutes approximately 70% of the total miRNA content in hepatic cells. miR-122 levels are considered more specific indicators of liver injury compared to traditional markers like ALT or AST. However, significant variability in miR-122 levels has been observed both among different individuals and within the same individual over time, even in

Table 6 Roussel Uclaf Causality Assessment Method (RUCAM) criteria [66]

Criterion	Description
Temporal association	Timing of liver injury about drug exposure
Improvement upon drug cessation	Liver injury improvement after stopping the drug
Risk factors	Presence of risk factors (e.g., age, alcohol use, preexisting liver disease)
Exclusion of other liver disorders	Ruling out alternative causes of liver injury
Known drug hepatotoxicity	Existing evidence of the drug causing liver injury
Recurrence on re-exposure	Liver injury recurrence upon re-exposure to the drug
Influence of other medications	Consideration of other drugs being taken

Table 7 Limitations of CIOMS/RUCAM Scale [67, 68]

Limitation	Description
Complexity	The scale is complex and difficult to apply
Ambiguous definitions	Some criteria have vague definitions
Lack of supporting data	Insufficient data supporting the validity of some criteria
Reliance on rechallenge data	High probability categories often require rechallenging data
Variability with preexisting liver Disease	Patients with preexisting liver disease may receive lower scores due to liver test fluctuations
Applicability to herbal and dietary supplements	Difficulty in applying the scale to herbal and dietary supplements due to ingredient variability

healthy adults. This variability may be attributed to the release of miR-122 from healthy liver cells, which can have physiological effects on distant tissues [71].

Glutamate dehydrogenase (GLDH) is a mitochondrial protein with less variability than miR-122 and is released into the bloodstream during liver cell damage, making it a promising biomarker for drug-induced liver injury (DILI). Both GLDH and miR-122 are FDA-endorsed for further study as liver-specific biomarkers. Other potential DILI biomarkers under investigation include high-mobility group protein B1 (HMGB-1), a nuclear protein released during necrosis, and the ratio of cleaved to full-length cytokeratin [72]. The comprehensive summary of molecular biomarkers is presented in (Fig. 4).

Prognosis

Drug-induced liver injury (DILI) usually improves once the medication causing it is stopped. However, it can sometimes progress to acute liver failure (ALF), leading to serious complications or the need for a liver transplant. The prognosis depends on factors, such as the drug involved, the severity of the injury, and the patient's characteristics. Overall, studies suggest that the outlook for DILI is better in the general population than in patients at specialized medical centers, where

cases are often more severe [73]. ALF is marked by significant changes in liver function, presenting symptoms such as encephalopathy, coagulopathy, and jaundice in individuals who do not have preexisting chronic liver disease. The mortality rate for ALF varies considerably, ranging from 60 to 90%, depending on factors like the underlying cause of liver failure and the expertise of the transplant center in managing liver transplants. This high mortality rate underscores the critical nature of prompt diagnosis and intervention in cases of ALF. Idiosyncratic DILI patients generally have a poorer prognosis [73, 74].

Several scoring systems have been developed to predict the prognosis of DILI. One significant contribution was made by Hyman Zimmerman in the 1960s, who observed that patients exhibiting hepatocellular DILI alongside jaundice had a notably elevated risk (between 10 and 50%) of progressing to acute liver failure—during a time when liver transplantation was not yet an option. This critical observation is now referred to as Hy's law, which is characterized by ALT level greater than three times the upper limit of normal (ULN) and total bilirubin exceeding two times the ULN in DILI patients, after other potential causes of liver injury have been ruled out (Fig. 5) [75].

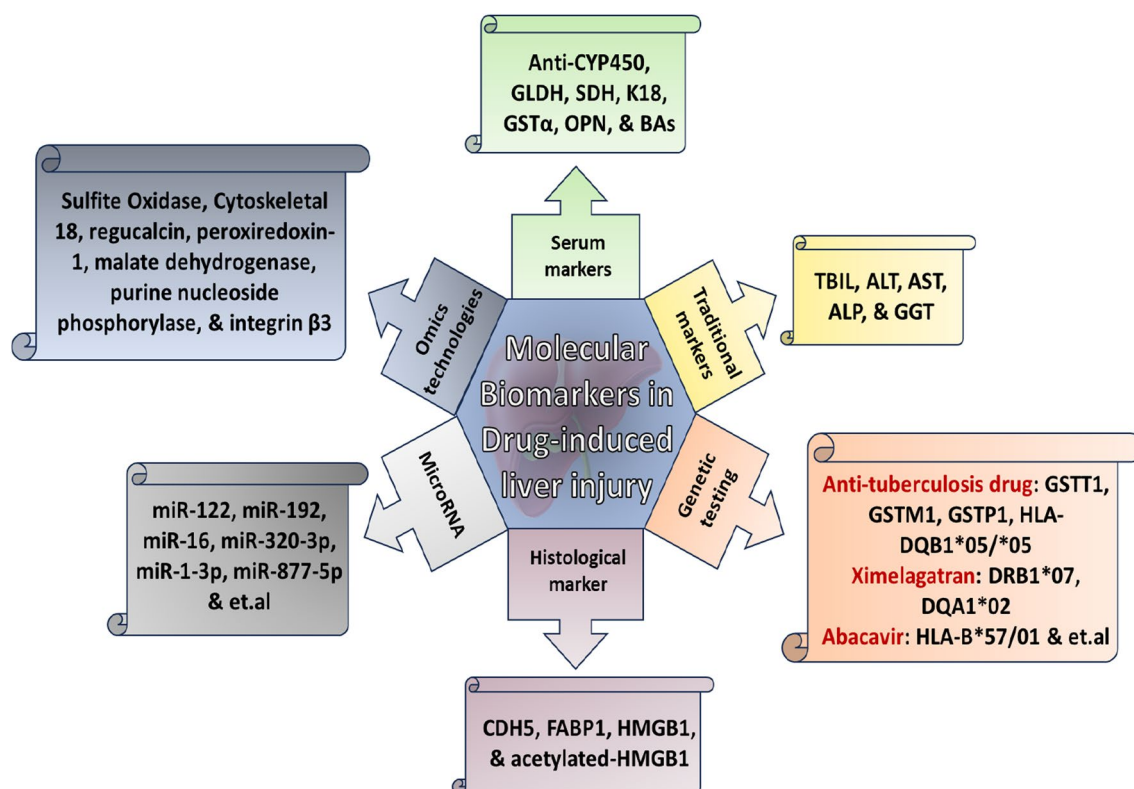


Fig. 4 Molecular biomarkers in drug-induced liver injury

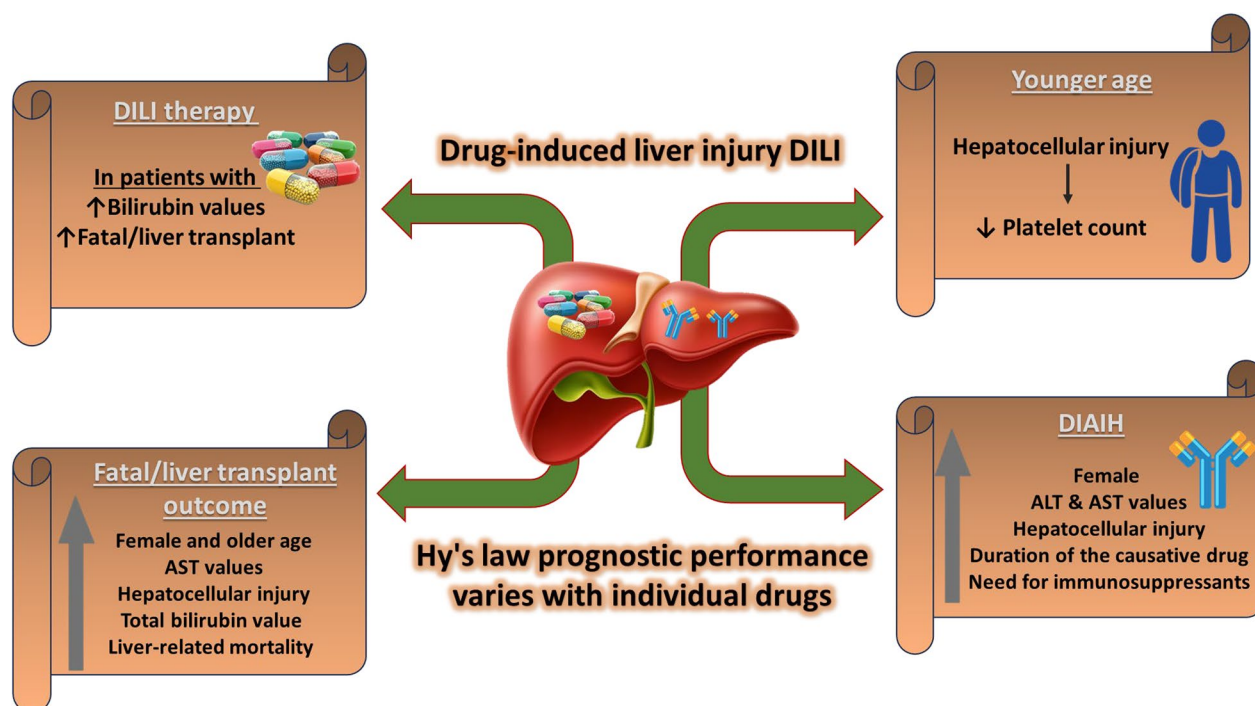


Fig. 5 Hy's law prognostic performance varies with individual drugs

Hy's law is used by the FDA to identify drugs that may cause severe liver injury during development. It originated from the observation that isoniazid-induced hepatocellular jaundice had a mortality rate over 10%. Subsequent studies have confirmed its effectiveness in predicting severe outcomes in drug-induced liver injury (DILI). Patients with hepatocellular jaundice who meet Hy's law criteria have fatality rates of 7–13%, while those with cholestatic jaundice have even higher fatality rates. For example, in the DILIN cohort, the fatality rate was around 14%, higher than in Swedish and Spanish DILI cohorts, where it ranged from 5 to 8% [76]. The prognosis for jaundice caused by drugs can vary significantly, with mortality rates ranging from 0% for erythromycin to 40% for halothane in cases of idiosyncratic drug-induced liver injury (DILI). Researchers from the Spanish hepatotoxicity network have worked to improve Hy's law by creating a more accurate model to predict acute liver failure (ALF) in DILI patients [77].

Most patients with DILI experience full recovery, with only a small proportion developing chronic DILI. Chronic DILI is defined by persistent liver biochemical abnormalities or imaging findings that last more than a year. In a study from a Spanish registry involving 292 DILI patients, 8% of cases progressed to chronic DILI, which included serious conditions such as liver cirrhosis and ductal lesions. Notably, there was no consistent

pattern of DILI linked to the progression toward chronicity, making long-term outcomes unpredictable in these patients [78].

Emerging risk factors for DILI

Susceptibility to DILI is influenced by multiple factors, including host (pharmacogenomic, age and gender, metabolic status, preexisting liver disease), drug characteristics, and environmental or extrinsic factors [79].

Host factor

Host factors, especially pharmacogenomic risk factors, play a crucial role in DILI development. Early candidate gene studies focused on drug-metabolizing genes and found genetic variants associated with different forms of DILI. However, recent genome-wide association studies (GWAS) have not consistently validated these links. Most pharmacogenomic risk factors for DILI reside in the HLA region on chromosome 6 and are often drug-specific. For example, variants in the HLA region are strongly associated with DILI from drugs like amoxicillin–clavulanate and flucloxacillin [80].

HLA screening is now used to help rule out DILI, thanks to its high negative predictive values for specific drugs. For instance, 34% of patients being considered for lumiracoxib would be excluded if they carry the HLA-DQA1*0102 allele, yet only 5.6% of these individuals are

expected to develop DILI. This demonstrates how genetic screening can reduce the risk of liver injury while minimizing unnecessary exclusions of potential patients [81]. However, HLA testing is particularly valuable for certain drugs like abacavir, as individuals with the HLA-B*5701 allele have a very high risk of developing severe hypersensitivity reactions, including DILI [82]. Another HLA risk allele identified in genome-wide association studies is the rs2476601 variant in the PTPN22 gene, which is associated with an increased risk of DILI from multiple drugs. PTPN22 plays a key role in regulating T cell responses, and its variants are linked to a higher susceptibility to autoimmune disorders, emphasizing the role of immune dysregulation in DILI mechanisms [80].

Age and gender Age and gender affect DILI susceptibility. Elderly individuals experience reduced hepatic function, increasing DILI risk due to slower metabolism and prolonged exposure. Conversely, developmental differences in children's liver enzyme activity may increase their DILI risk [83]. Women may be more susceptible to DILI than men, possibly due to hormonal influences on liver enzyme activity and drug metabolism. Understanding

these gender differences is crucial for tailoring prevention and treatment strategies [84].

Metabolic effect by cytochrome P450 It was observed that drugs that undergo extensive metabolism by CYP450 enzymes are generally considered more likely to cause DILI compared to drugs that are not metabolized. The reactive metabolites generated after drug biotransformation can covalently bind to cellular proteins, lipids, and nucleic acids, leading to triggering immune responses and cellular damage (Fig. 6).

Preexisting disease The preexisting liver disease significantly impacts susceptibility to DILI in several ways. Liver diseases, such as cirrhosis, can reduce metabolic capacity, impairing drug metabolism and clearance. This can lead to drug or metabolite accumulation due to reduced hepatic function, altered enzyme activity, and impaired bile flow, thereby increasing the risk of DILI [85]. Additionally, liver diseases can make hepatocytes more sensitive to injury and compromise the liver's regenerative capacity, increasing vulnerability to drug-induced hepatotoxicity. Altered immune responses in these conditions may impair the recognition and clearance of drug-induced antigens or

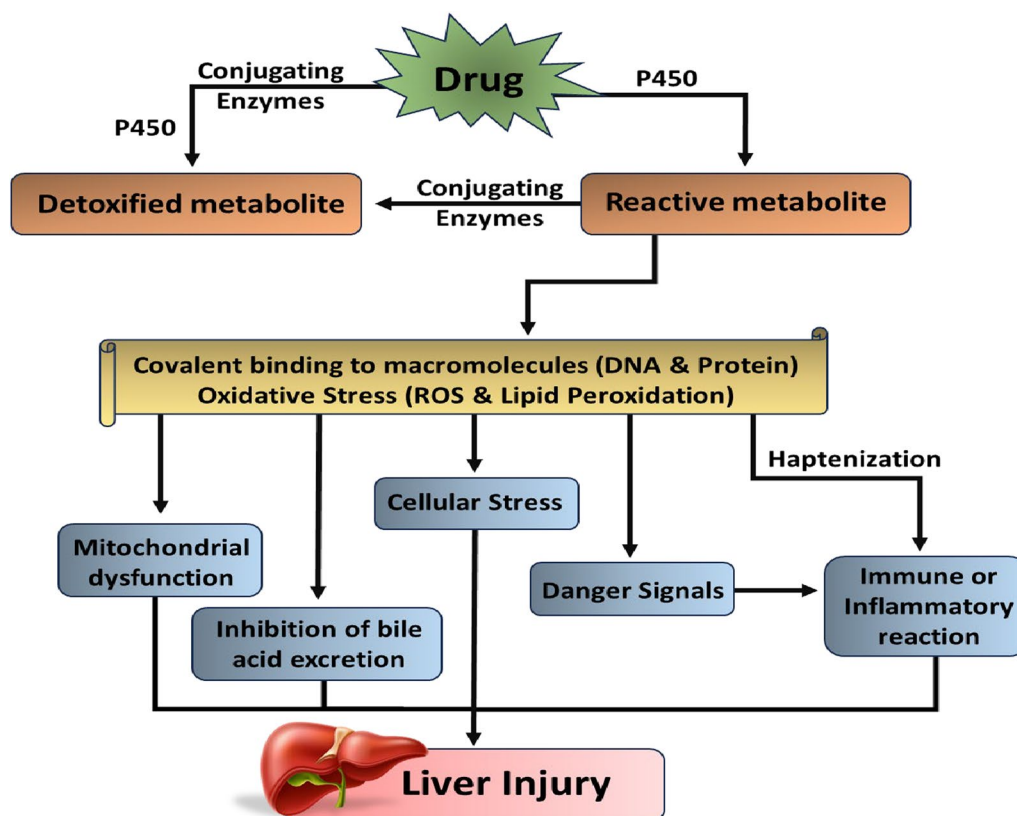


Fig. 6 Mechanisms of drug-induced liver injury associated with metabolic effect by cytochrome P450 on drugs

immune complexes, contributing to immune-mediated DILI. Additionally, impaired bile flow and altered drug metabolism in hepatic dysfunction can cause drug accumulation in the liver, further elevating the risk of toxicity [86]. Managing drug-induced liver injury (DILI) in patients with preexisting liver conditions is challenging due to overlapping symptoms and abnormal liver enzyme levels. Exposure to harmful substances can worsen existing liver problems, particularly with the rise of obesity, poor diets, and inactive lifestyles. Metabolic liver diseases and xenobiotic exposure can cause similar cellular stress, worsening liver health and potentially speeding up disease progression. Chronic liver conditions, such as NAFLD and alcoholic liver disease, also reduce the liver's ability to regenerate [87].

Drug features Drug characteristics play a main role in the risk of DILI, influenced by factors such as chemical structure, molecular weight, and lipophilicity. High lipophilicity ($\text{Log}P \geq 3$) increases a drug's permeability and uptake, enhancing biotransformation and potentially leading to higher levels of toxic metabolites. Furthermore, drugs with higher lipophilicity are enriched among those with age-biased DILI reporting frequency [88]. Compounds

metabolized by CYP450 enzymes, especially those producing reactive metabolites, are linked to cellular stress and a heightened risk of idiosyncratic DILI. Recent efforts by drug companies include screening candidate drugs for their potential for covalent binding, which is associated with DILI incidence. While some drugs induce DILI through reactive metabolites, others can do so through different mechanisms [89]. The research by Teschke and Danan showed that most drugs causing a specific type of liver injury are broken down by enzymes called CYP450s. Notably, nearly half are processed by CYP3A4 and CYP3A5 specifically, with another quarter handled by CYP2C8 or CYP2C9. Conversely, drugs containing a carboxylic acid functional group may pose a lower risk of DILI, as they are typically metabolized to acyl glucuronides that are readily excreted (Fig. 7) [90].

Other emergency factors Identifying emergency risk factors for DILI is crucial as these can exacerbate the condition and lead to more severe liver damage. One significant risk factor is polypharmacy, which involves the concurrent use of multiple medications [83]. This increases the risk of DILI due to potential drug–drug interactions and the cumulative hepatotoxic effects of several substances.

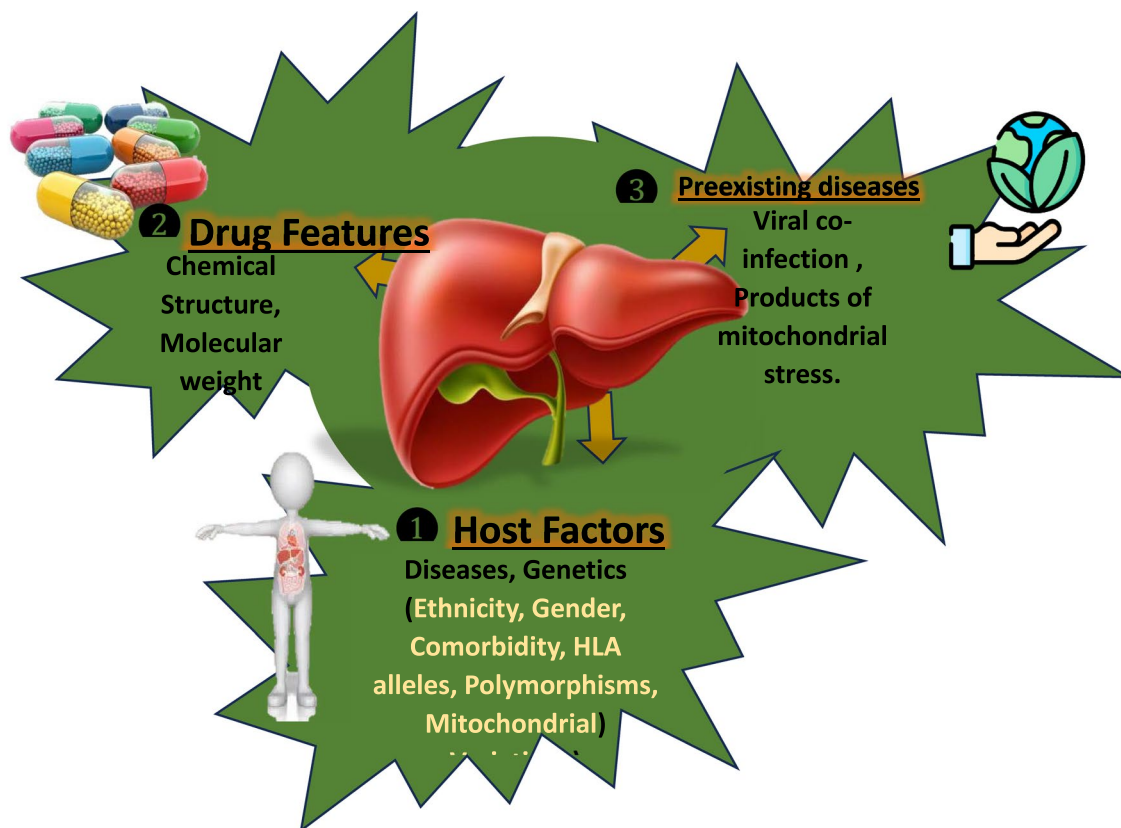


Fig. 7 Emerging risk factors for DILI

Patients on polypharmacy require careful monitoring for liver function and potential signs of liver injury [91].

Genetic variations can affect drug metabolism, leading to an induced risk of hepatotoxicity. Polymorphisms in genes encoding drug-metabolizing enzymes, such as CYP450, can alter the way drugs are processed in the liver, making genetic screening a potential tool for identifying individuals at higher risk for DILI [92]. Nutritional status also plays a main role in the risk of DILI. Malnutrition or obesity can affect liver function and drug metabolism. Malnutrition may impair liver regeneration, while obesity is often associated with NAFLD [93]. Nutritional interventions may be crucial for optimizing liver health and reducing DILI risk. Underlying health conditions like diabetes, renal impairment, and cardiovascular diseases can increase the likelihood of DILI, making comprehensive management of these conditions essential to minimize liver stress [94].

Common drugs associated with DILI

Old described hepatotoxic agents DILI can be triggered by a variety of medications, including both prescription and OTC drugs. Antibiotics are a major contributor to DILI because of their broad usage and diverse mechanisms of action. For example, amoxicillin–clavulanate, isoniazid, and fluoroquinolones are known to induce hepatocellular, cholestatic, or mixed types of liver injury [95]. NSAIDs are also commonly associated with liver damage, particularly with long-term use. Diclofenac, ibuprofen, and naproxen are examples of NSAIDs that can cause hepatocellular injury, often linked to mitochondrial dysfunction and oxidative stress [62].

Acetaminophen is a common antipyretic and analgesic, but overdose can lead to severe liver damage and acute liver failure due to the toxic metabolite NAPQI,

which depletes glutathione and causes oxidative stress. Additionally, statins used to lower cholesterol levels can elevate liver enzymes and, in rare cases, cause severe liver injury. Atorvastatin, simvastatin, and rosuvastatin are examples associated with these risks, with potential mechanisms of injury including immune-mediated responses or mitochondrial dysfunction [96].

Antiepileptic drugs are essential for managing epilepsy but carry a risk of liver injury. Examples include valproic acid, carbamazepine, and phenytoin. Hepatocellular injury is common with these drugs, with mechanisms including metabolic activation and immune-mediated damage. Antituberculosis medications, such as isoniazid, pyrazinamide, and rifampicin, can cause significant liver toxicity, with hepatocellular injury often resulting from reactive metabolite formation and immune responses [97].

Immunosuppressants used in organ transplantation and autoimmune diseases can cause liver injury. Examples include methotrexate, azathioprine, and tacrolimus [98]. These medications can cause steatosis, fibrosis, and hepatocellular injury through various mechanisms, including direct toxicity and immune modulation. Herbal and dietary supplements are not immune to causing liver injury, often due to contaminants, adulterants, or intrinsic hepatotoxicity. Examples include green tea extract, kava, and black cohosh, with patterns of injury varying widely, including cholestatic, hepatocellular, and mixed types (Table 8) [99].

Newly described hepatotoxic agents Recent advancements in pharmacology and toxicology have led to the identification of several new hepatotoxins. These substances are of significant concern due to their potential impact on public health. Understanding these hepato-

Table 8 Old described hepatotoxic agents

Drug class	Examples	Hepatotoxic effects	Mechanisms
Antibiotics	Amoxicillin–clavulanate, isoniazid, fluoroquinolones	Hepatocellular, cholestatic, mixed	Varies by drug (e.g., reactive metabolites, immune-mediated)
NSAIDs	Diclofenac, ibuprofen, naproxen	Hepatocellular	Oxidative stress, mitochondrial dysfunction
Analgesics/antipyretics	Acetaminophen	Severe liver injury, acute liver failure	NAPQI production, oxidative damage
Statins	Atorvastatin, simvastatin, rosuvastatin	Liver enzyme elevations, severe injury	Immune-mediated, mitochondrial dysfunction
Antiepileptics	Valproic acid, carbamazepine, phenytoin	Hepatocellular	Metabolic activation, immune-mediated damage
Antituberculosis	Isoniazid, rifampicin, pyrazinamide	Significant liver toxicity	Reactive metabolites, immune responses
Immunosuppressants	Methotrexate, azathioprine, tacrolimus	Steatosis, fibrosis, hepatocellular	Direct toxicity, immune modulation
Herbal/dietary supplements	Green tea extract, kava, black cohosh	Hepatocellular, cholestatic, mixed	Contaminants, adulterants, intrinsic hepatotoxicity

toxins is crucial for developing preventive strategies and therapeutic interventions [100]. Malnick et al. described a case of submassive hepatic necrosis caused by denosumab, which was linked to an immune reaction to the monoclonal antibody. In this case, the patient's aminotransferase and gamma-glutamyl transferase (GGT) levels improved after treatment with steroids [101]. Namn et al. documented the first reported case of acute liver failure associated with diphenhydramine in a 28-year-old man who was taking 400 mg nightly for sleep, without any concurrent use of acetaminophen. It is noteworthy that neither the RUCAM nor any other causality scoring system was utilized in this instance [102]. Nijima et al. reported a case involving a 75-year-old patient with type 2 diabetes mellitus who experienced cholestatic DILI following a switch to oral ipragliflozin, a sodium-glucose cotransporter-2 inhibitor. The patient's condition improved after discontinuing the medication [103].

Studies have shown that tolvaptan, a vasopressin V2 receptor antagonist used for treating hyponatremia, is associated with a higher incidence of liver injury indicated by elevated liver enzymes compared to placebo. Some cases meet Hy's Law criteria for severe liver injury, typically developing after 3–18 months of use and improving upon discontinuation. While no cases of acute liver failure (ALF) were reported, there is a very low estimated risk of 1 in 4,000. As a result, regular monitoring of liver enzymes (ALT) is recommended for patients on long-term tolvaptan therapy [104].

Russmann et al. reported 14 cases of liver injury linked to rivaroxaban, with two of these cases confirmed through liver biopsy. Following this, other researchers have begun to investigate the hepatotoxicity associated with novel oral anticoagulants (NOACs), expanding the understanding of their potential impact on liver health [105]. In early 2018, the FDA released a safety statement regarding obeticholic acid, used for treating primary biliary cirrhosis, highlighting concerns over liver injury associated with its use. The agency noted that several patients were prescribed supratherapeutic doses, while some instances of liver injury occurred even at appropriate doses. The FDA underscored the importance of closely monitoring liver enzyme levels in patients prescribed this medication to mitigate risks [106].

Ecuzumab (Soliris), used for atypical hemolytic uremic syndrome (aHUS) and paroxysmal nocturnal hemoglobinuria (PNH), may cause drug-induced liver injury (DILI). Clinicians should monitor liver function, as elevated liver enzymes have been reported in patients. A study noted elevated aminotransferases in seven of 11 children treated for aHUS, with one case of severe elevation leading to treatment discontinuation. Despite not being required by the drug's Risk Evaluation and

Mitigation Strategy (REMS), regular liver enzyme monitoring is recommended [107].

Ipilimumab, a monoclonal antibody targeting CTLA-4 for advanced melanoma, can cause immune-related adverse events (irAEs), including colitis and hepatitis. These irAEs usually appear 3–9 weeks after treatment initiation and often resolve within two weeks with appropriate management. Cheng et al. reported three male patients who developed hepatitis due to ipilimumab, all of whom improved and showed decreased ALT levels following high-dose intravenous prednisolone (1 g daily for 3–5 days) [108].

Solithromycin is a next-generation macrolide antibiotic, which addresses macrolide resistance in community-acquired bacterial pneumonia (CABP). Unlike traditional macrolides, it binds to the 50S ribosomal subunit of bacteria with a similar mechanism of inhibiting protein synthesis [109].

Solithromycin's longer alkyl-aryl side chains improve its binding to the ribosome, enhancing its effectiveness against macrolide-resistant strains. During Phase III clinical trials, increased liver enzymes indicated potential hepatotoxicity, with most patients reaching peak ALT levels within four days of starting treatment. However, up to 30% of patients may experience elevated liver enzymes later, even up to 15 days after discontinuation, with this delay occurring more frequently with the intravenous formulation compared to the oral version [110].

Accordingly, it was not approved by the FDA due to this concern. While numerous, the FDA felt the existing clinical trials did not involve enough participants to definitively assess the risk. They requested a large safety study with at least 9,000 patients before considering approval to address this problem. The summary of newly described hepatotoxic agents is presented in (Table 9). The comparison between mechanisms of liver injury induced by old and new drugs is presented in (Fig. 8).

Emerging findings on well-known hepatotoxins

APAP Although there is a substantial body of research focused on acetaminophen-induced DILI, there have been a relatively small number of recent publications that have notably enhanced our understanding of other intrinsic hepatotoxins. Recent advances in APAP hepatotoxicity research have focused on several areas such as novel diagnostic biomarkers, genetic susceptibility to APAP hepatotoxicity, new treatment strategies, and public health initiatives [115].

Jiang et al. investigated the use of acetaminophen protein adducts as possible biomarkers for diagnosing liver injury caused by acetaminophen in both children and adults. Their cross-sectional study, which included 100 children aged 1 to 12 years, revealed that 52% of those

Table 9 Newly described hepatotoxic agents and their clinical implications

Drug	Condition treated	Hepatotoxic effects	Key findings and cases
Denosumab	Osteoporosis	Submassive hepatic necrosis	Improved aminotransferase and GGT levels after steroid treatment [100, 111]
Diphenhydramine	Allergies, sleep aid	Acute liver failure	First known case without acetaminophen; no RUCAM score applied [102]
Ipragliflozin	Type 2 diabetes mellitus	Cholestatic DILI	Improved upon discontinuation; being studied for NAFLD treatment [103]
Tolvaptan	Hyponatremia	Elevated liver enzymes, risk of liver injury (Hy's law)	Regular ALT monitoring is advised; no cases of acute liver failure observed [104]
Rivaroxaban	Anticoagulation	Liver injury	14 cases reported; DILI accounted for 3.7% of adverse event reports [112]
Obeticholic acid	Primary biliary cirrhosis	Liver injury linked to inappropriate dosing	Emphasis on monitoring liver enzymes closely [106]
Benzodiazepines	Anxiety, insomnia	Hepatotoxicity concerns	High-dose use did not lead to significant hepatotoxicity in the study [113]
Ecuzimab	aHUS, PNH	Elevated aminotransferases, liver injury	Monitoring liver enzymes recommended despite not being in REMS [114]
Ipilimumab	Advanced melanoma	Immune-related hepatitis	Symptoms improved with high-dose steroid regimen [108]
Solithromycin	Bacterial infections	Elevated liver enzymes	Delayed ALT elevations noted; FDA requested large safety study before approval [110]

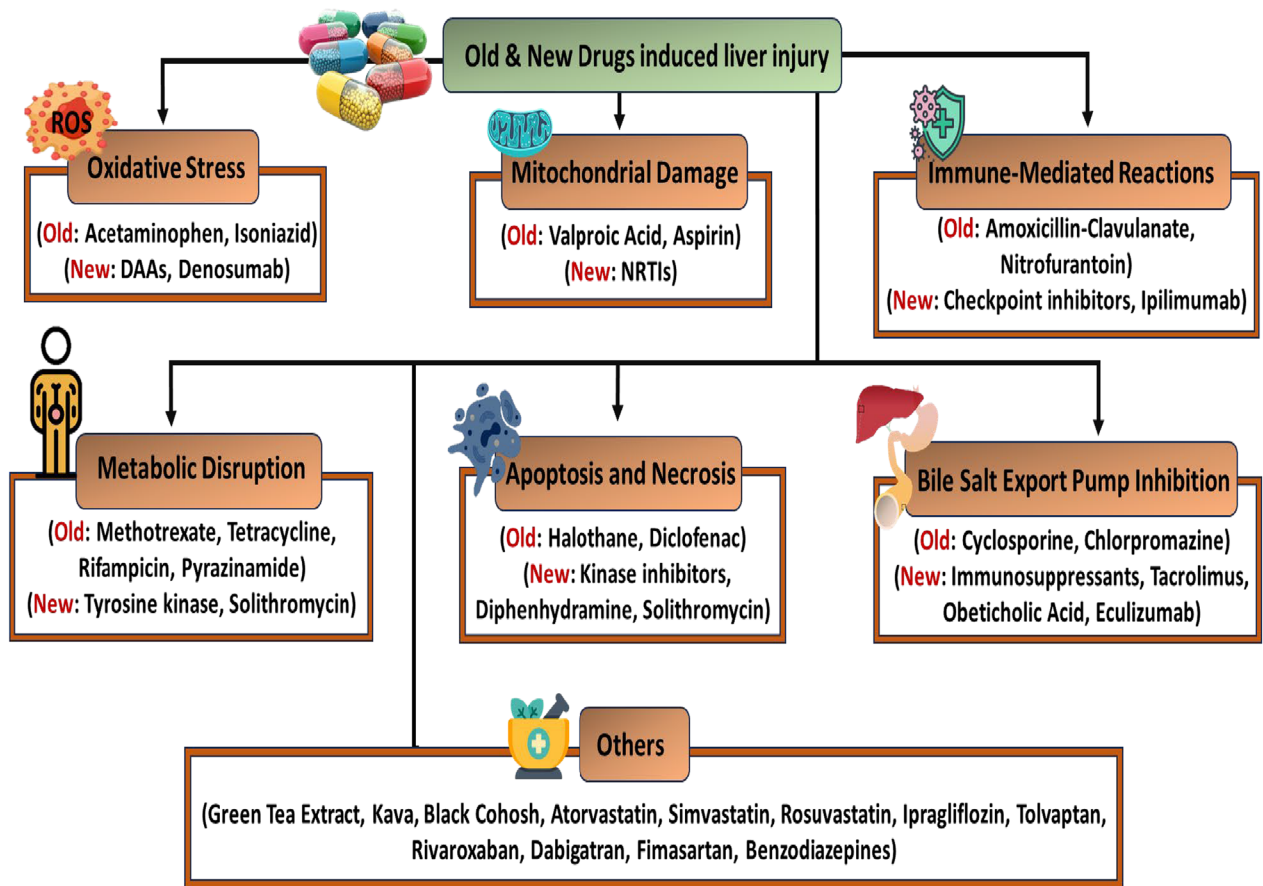


Fig. 8 Mechanisms of Liver Injury by Old and New Drugs. ROS Reactive oxygen species; DAAs Direct-acting antivirals; NRTIs Nucleos(t)ide reverse transcriptase inhibitors

who had taken acetaminophen within the last two weeks tested positive for the adducts. In contrast, all participants who reported no recent acetaminophen use tested negative [116]. In a comparable study with 230 adults, it was found that 42% of individuals who indicated recent acetaminophen consumption tested positive for the adducts. Conversely, 99% of those who denied recent use tested negative [117]. Both studies indicated a positive relationship between the serum levels of acetaminophen protein adducts and the timing and dosage of acetaminophen intake. Additionally, several serum biomarkers, such as microRNA-122, keratin-18 (K-18), and high-mobility group box-1 (HMGB-1), are associated with the mechanisms underlying DILI in humans. Notably, markers related to mitochondrial damage appear to have the potential to predict the outcomes of acetaminophen-induced toxicity [116, 117].

Antimicrobials Isoniazid is a main cause of DILI and even ALF. Current recommendations often rely on patients reporting symptoms like hepatitis instead of regular monitoring of liver enzymes (ALT tests) [118]. Studies suggest that isoniazid-induced DILI is under-reported, with the US DILI network finding it as the second most common risk. A worrying trend is the delay in stopping isoniazid after symptom onset. Over half of patients with concerning symptoms (nausea, fatigue, etc.) continued taking the drug for over a week, and 40% for more than two weeks. Continuing isoniazid despite symptoms significantly worsened outcomes [118, 119].

Four macrolide antibiotics are available in the US (erythromycin, azithromycin, telithromycin, and clarithromycin), and all have been linked to liver injury of varying severity [120]. While traditionally, acute cholestasis with erythromycin was thought to be self-limited, recent DILI network data suggest a potential for chronic DILI. Notably, azithromycin, more commonly prescribed than others, appears to cause chronic DILI in over 20% of cases, particularly in patients with underlying liver disease [121]. Most cases of cefazolin-induced DILI were classified as moderate to severe, although they were typically self-limiting [122].

Enalapril Enalapril, a medication classified as an angiotensin-converting enzyme (ACE) inhibitor, is commonly prescribed for managing high blood pressure and heart failure. Although it is generally well-tolerated, enalapril can, in rare instances, lead to liver injury [123]. Vigibase, the World Health Organization's database of medication safety reports, contains documented cases of enalapril-induced liver injury [124]. Hendriks et al. proposed a hypothesis in a mouse model, suggesting oxidative stress in livers with preexisting fatty liver disease might play a

role. While this requires confirmation in human studies, such findings could inform clinical decisions regarding ACE inhibitor prescription for patients with preexisting steatosis [125].

Management

The cornerstone of DILI management lies in early identification and prompt withdrawal of the offending agent. Delaying this withdrawal can significantly increase the risk of progression to ALF and necessitate liver transplantation. Re-introducing the offending agent is generally discouraged unless essential in a clinical setting. Meticulous monitoring of liver function tests is essential [126]. While treatment options for DILI affecting liver cells (hepatocellular DILI) are limited, corticosteroids are often used for severe cases of liver dysfunction. However, their effectiveness lacks strong evidence, with exceptions for cases where autoimmune hepatitis (AIH) cannot be ruled out or for DILI caused by immune-checkpoint inhibitors (ICIs) that are used in the treatment of cancer (Fig. 9).

Medications, like L-carnitine and N-acetylcysteine, may be used in specific cases to address underlying causes (intrinsic DILI), but there is no universal antidote for DILI. Treatment focuses on supporting the liver's recovery by promoting detoxification, eliminating free radicals with antioxidants, and protecting liver cells. A promising study examined the use of NAC for treating non-acetaminophen-induced ALF, including cases of DILI. This randomized, placebo-controlled trial revealed that patients with this form of ALF who were treated with NAC had a significantly higher likelihood of surviving without the need for a liver transplant compared to those who did not receive NAC [127]. For managing symptoms of severe itching in patients with cholestatic DILI, antihistamines like hydroxyzine or diphenhydramine, or bile acid resins like cholestyramine, can be helpful. Ursodeoxycholic acid is another medication sometimes used off-label for this condition. Following identification and withdrawal of the offending medication, most patients with DILI experience gradual improvement in clinical symptoms and blood test results over days or weeks. However, close monitoring remains crucial as some cases can deteriorate [128].

The decision to hospitalize a patient or determine the duration of hospitalization is based on the severity of liver dysfunction. Patients with jaundice, particularly those with hepatocellular DILI, who are at a higher risk of progressing to acute liver failure, typically need close monitoring and are likely to be hospitalized [129]. The development of coagulopathy, early signs of mental confusion, and impaired kidney function, which can indicate impending liver failure, necessitates a referral

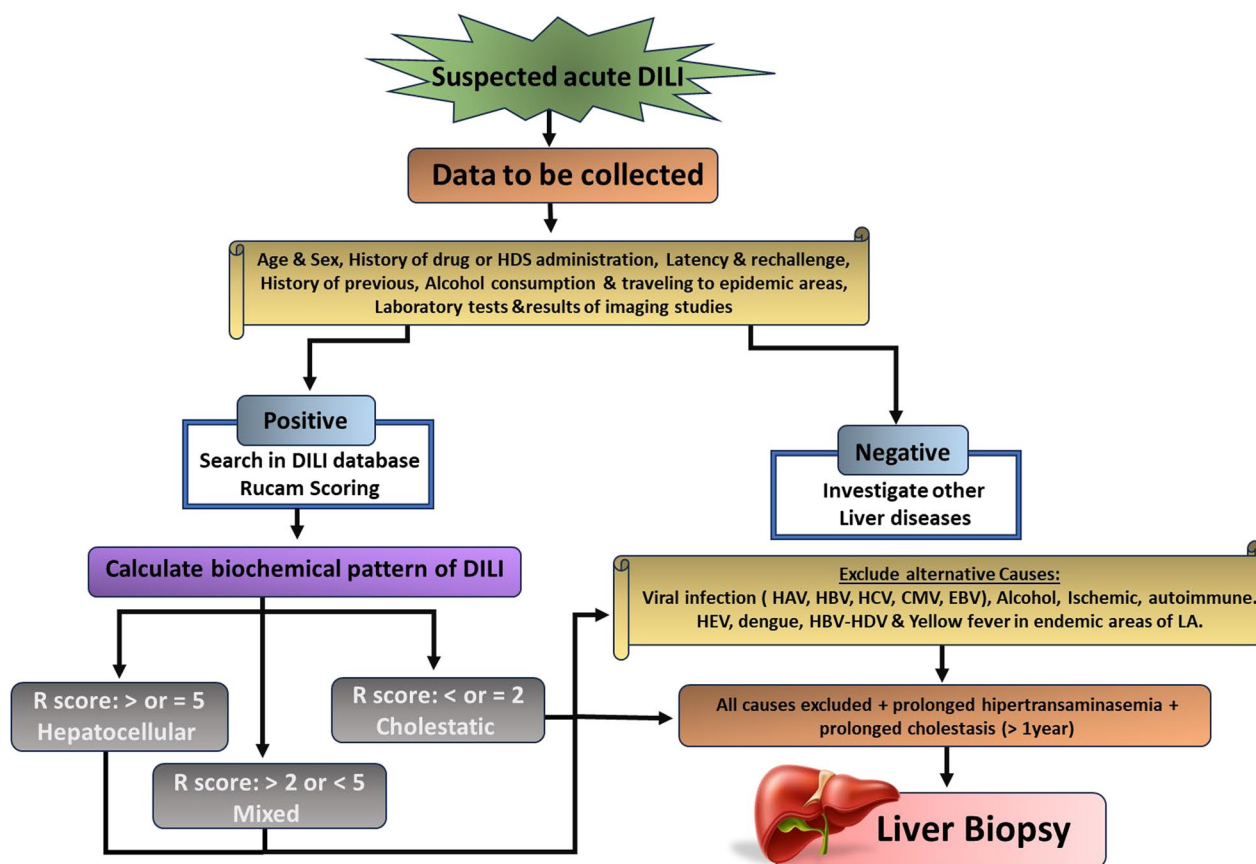


Fig. 9 The management strategy for DILI

immediately to a liver transplant center for evaluation. Additionally, for patients with DILI specifically affecting liver cells (hepatocellular DILI), progressively worsening jaundice should also raise concerns and prompt evaluation for a potential liver transplant by a specialized center [130].

Conclusion

DILI remains a critical challenge in clinical practice and drug development. Understanding the distinct mechanisms underlying intrinsic and idiosyncratic DILI is essential for improving diagnosis, management, and prevention strategies. Despite advances in research, the unpredictable nature of idiosyncratic DILI and the complexities of individual susceptibility continue to pose significant obstacles. Enhanced pharmacovigilance, genetic screening, and personalized medicine approaches may hold the key to mitigating the impact of DILI. Future research should aim to clarify the specific molecular pathways involved, establish dependable biomarkers for early detection, and investigate new therapeutic targets to enhance patient outcomes.

Abbreviations

AIH	Autoimmune hepatitis
ALF	Acute liver failure
ALT	Alanine aminotransferase
APAF-1	Apoptotic protease activating factor-1
APAP	Acetaminophen (N-acetyl-p-aminophenol)
AREs	Antioxidant response elements
ASH	Alcoholic steatohepatitis
ASMA	Anti-smooth muscle antibodies
AST	Aspartate aminotransferase
ATP	Adenosine triphosphate
BAs	Bile acids
Bcl-2	B-cell lymphoma 2
BSEP	Bile salt export pump
CABP	Community-acquired bacterial pneumonia
CD8+	Cluster of differentiation 8 positive
CD20+	Cluster of differentiation 20 positive
CIOMS	The council for international organizations of medical sciences
CT	Computed tomography
CTLA-4	Cytotoxic T-lymphocyte-associated protein 4
CXCL10	C-X-C motif chemokine ligand 10
CYP2C8	Cytochrome P450 2C8
CYP2C9	Cytochrome P450 2C9
CYP3A4	Cytochrome P450 3A4
CYP3A5	Cytochrome P450 3A5
CYP450	Cytochrome P450
DAMPs	Damage-associated molecular patterns
DAAs	Direct-acting antivirals
DDW-J	Digestive disease week Japan
DILI	Drug-induced liver injury

DNA	Deoxyribonucleic acid
DR5	Death receptor 5
ER	Endoplasmic reticulum
Fas	Fas cell surface death receptor
FasL	Fas ligand
FADD	FAS-associated death domain protein
GR	Glutathione reductase
GSH	Glutathione
GGT	GGT: Gamma-glutamyl transferase
GWAS	GWAS: Genome-wide association studies
HAV	Hepatitis A virus
HBV	Hepatitis B virus
HCV	Hepatitis C virus
HDS	Herbal and dietary supplements
HEV	Hepatitis E virus
HLA	Human leukocyte antigens
HMGB-1	High-mobility group box-1
HO-1	Heme oxygenase-1
HPLC	High-performance liquid chromatography
HSV	Herpes simplex virus
IgG	Immunoglobulin G
IL-1 β	Interleukin 1 beta
IL-6	Interleukin 6
IL-8	Interleukin 8
ICI	Immune checkpoint inhibitor
IRE1 α	Inositol-requiring enzyme 1 alpha
JNK	C-Jun N-terminal kinase
Keap1	Kelch-like ECH-associated protein 1
LC3	Microtubule-associated protein 1 light chain 3
LogP	Logarithm of the partition coefficient
LTT	Lymphocyte transformation test
MAPK	Mitogen-activated protein kinases
MDR	Multidrug resistance
mPTP	Mitochondrial permeability transition pore
MRCP	Magnetic resonance cholangiopancreatography
MRP2	Multidrug resistance-associated protein 2
mRNA	Messenger ribonucleic acid
NASH	Nonalcoholic steatohepatitis
NAPQI	N-acetyl-p-benzoquinone imine
NAFLD	Nonalcoholic fatty liver disease
NF- κ B	Nuclear factor kappa B
NKT	Natural killer T
NOACs	Novel oral anticoagulants
NSAIDs	Nonsteroidal anti-inflammatory drugs
Nrf2	Nuclear factor erythroid 2-related factor 2
OAT	Organic anion transporter
OATP	Organic anion transporting polypeptide
OCT	Organic cation transporter
p53	Tumor protein 53
PERK	Protein kinase R-like endoplasmic reticulum kinase
PNH	Paroxysmal nocturnal hemoglobinuria
PTPN22	Protein tyrosine phosphatase non-receptor type 22
ROS	Reactive oxygen species
RUCAM	Roussel Uclaf Causality Assessment Method
SDH	Succinate dehydrogenase
TNF- α	Tumor necrosis factor alpha
TRAIL	TNF-related apoptosis-inducing ligand
ULN	Upper limit of normal
UPR	Unfolded protein response

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Author contributions

Ahmed K. Saleh was involved in conceptualization, methodology, writing—draft and editing, writing—final review. Nageh A. El-Mahdy helped in visualization, review, and supervision. Thanaa A. El-Masry contributed to visualization, review, and supervision. Nada A. Ashour was involved in literature search, collection, design draft preparation. Aya H. El-kadem helped in visualization, review, and supervision.

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